COURSE OUTLINE

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Certificate Renewal Automation with Claude

From expiring certificate to renewed, validated and audited binding on every load balancer

3 days (8 hours per day) | Intermediate | Instructor-led, hands-on labs | Capstone project

About this course

Who should attend

Enterprise IT and network operations teams, load balancer and ADC engineers, PKI and certificate owners, network automation leads, in India and worldwide.

Prerequisites

- Working knowledge of at least one load balancer platform (F5, NetScaler, Radware or similar)
- TLS basics: keys, certificates, chains, SANs and expiry
- Basic Python and Git
- Familiarity with an AI coding assistant is helpful, not required

What you will be able to do

- Explain the certificate lifecycle as each load balancer platform implements it and where renewal outages really come from
- Set up Claude Code, the Claude Agent SDK and an MCP server as a safe workbench for certificate operations
- Build a typed certificate inventory from F5 iControl REST, NetScaler NITRO, Radware Alteon and other load balancers (Anaya LB)
- Use a Claude planning agent to draft renewal plans as structured JSON and open ServiceNow change records with citations to inventory evidence
- Write deterministic, pytest-covered executors that upload, install, bind and verify certificates per platform
- Enforce guardrails, human approval, pre- and post-checks, automatic rollback and an audit trail reviewable through GitHub pull requests and Actions

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Tools and technologies

Claude Code | Claude Agent SDK and MCP | Python | F5 BIG-IP iControl REST / tmsh | Citrix NetScaler NITRO API | Radware Alteon | ACME lab CA and TLS certificates | ServiceNow Table API / Flow Designer | pytest | GitHub | GitHub Actions

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MODULE 1

Certificate lifecycle on load balancers and the automation baseline

- How each platform stores and binds certificates: F5 client-ssl profiles and virtual servers, NetScaler sslcertkey and sslvserver bindings, Radware Alteon certificate repository and SSL policies, other load balancers (Anaya LB)
- Where renewal outages come from: wrong chain, mismatched key, binding missed on a standby, HA sync skipped
- The operating pattern for the whole course: Claude plans, drafts and explains; deterministic code executes; a human approves; everything is logged
- Lab estate walkthrough: lb-f5-01, lb-ns-01, lb-rw-01, an ACME lab CA and a ServiceNow developer instance

HANDS-ON LAB
Renew one certificate by hand on the F5 and the NetScaler lab devices, record every step and every point where a mistake could cause an outage.

DELIVERABLE
Manual renewal runbook and failure-point map.

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MODULE 2

Setting up Claude Code, the Agent SDK and MCP for certificate work

- Claude Code project setup: CLAUDE.md with estate conventions, naming rules for certificate objects, and what the agent may never do
- Claude Agent SDK in Python: defining tools with JSON schemas, forcing structured JSON outputs, handling tool-use loops
- An MCP server exposing read-only inventory tools (list_certs, get_expiry, get_bindings) and no write tools
- Secrets handling: device credentials in environment variables and a vault, never in prompts or repositories

HANDS-ON LAB
Build the read-only certificate MCP server, connect it to Claude Code and ask Claude to explain the bindings on lb-f5-01 without changing anything.

DELIVERABLE
Working agent workbench with read-only MCP tools.

MODULE 3

Certificate inventory and expiry discovery across platforms

- F5 iControl REST: /mgmt/tm/sys/file/ssl-cert, /mgmt/tm/ltm/profile/client-ssl and virtual server lookups; tmsh equivalents for verification
- NetScaler NITRO: /nitro/v1/config/sslcertkey and sslvserver_sslcertkey_binding; Radware Alteon REST API certificate and SSL policy objects; an adapter interface for other load balancers (Anaya LB)
- A typed inventory record: device, platform, certificate name, subject and SANs, chain, expiry, bound virtual servers, owner, confidence
- Claude drafts the collectors; you review the diffs and run pytest against recorded API responses

HANDS-ON LAB
Build the multi-platform inventory collector and produce a single expiry report for the lab estate with unknowns marked explicitly.

DELIVERABLE
Certificate inventory and expiry report.

DAY 2 3 modules

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MODULE 4

Renewal planning agent and ServiceNow change records

- Planning agent: group certificates by expiry window, owner and platform; propose order, maintenance window and blast radius as structured JSON
- Mandatory citations to inventory rows; unknowns (missing owner, unverified binding) block the plan instead of being guessed
- Opening and updating change_request via the ServiceNow Table API; triggering an approval path in Flow Designer
- Requesting the new certificate from the ACME lab CA or an internal CA request with CSR generated by code, not by the model

HANDS-ON LAB

Run the planning agent on the inventory, inspect the JSON plan, and watch it open a change record that cites the exact certificates it will touch.

DELIVERABLE

Renewal planning agent and change-record integration.

MODULE 5

Deterministic executor: F5 BIG-IP with iControl REST and tmsh

- Upload key and certificate through /mgmt/shared/file-transfer/uploads, then create sys file ssl-key and ssl-cert objects
- Update the client-ssl profile cert-key-chain and confirm virtual server bindings; verify with tmsh list ltm profile client-ssl
- Idempotency, naming with date suffixes, and config sync across the HA pair before declaring success
- pytest with recorded iControl responses; Claude drafts code, explains each call, and never holds the credentials

HANDS-ON LAB

Build and test the F5 executor against a mock iControl endpoint, then run it against lb-f5-01 in dry-run and live modes.

DELIVERABLE

Tested F5 certificate executor.

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MODULE 6

Deterministic executors: NetScaler NITRO, Radware Alteon and other load balancers (Anaya LB)

- NITRO API: upload with systemfile, add or update sslcertkey, bind to sslvserver, save config; handling HA primary-only writes
- Radware Alteon: certificate import and SSL policy update through the Alteon REST API, with apply and save as explicit steps
- Adapter pattern for other load balancers (Anaya LB): one interface, one contract test suite, platform-specific implementation
- A common executor contract: plan in, result and evidence out, every call logged with request and response IDs

HANDS-ON LAB

Implement the NetScaler and Radware executors, prove all three platforms pass the same contract tests, and stub the adapter for other load balancers (Anaya LB).

DELIVERABLE

Multi-platform executor library with contract tests.

DAY 3 3 modules

MODULE 7

Guardrails, approvals and audit for certificate changes

- Policy checks before any write: approved change in ServiceNow, maintenance window open, device on allow list, HA healthy and in sync
- Human approval gate: the approver sees the plan, the diff of bindings and the rollback command; the agent cannot self-approve
- Scope limits: the model proposes, executors refuse anything outside the plan, and credentials never pass through Claude
- Audit log of prompts, tool calls, device responses, approver and timestamps; evidence attached back to the change record

HANDS-ON LAB

Attempt a renewal without an approved change and outside the window, and watch each guardrail block it with a logged reason.

DELIVERABLE

Guardrail policy module and audit trail.

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MODULE 8

Pre- and post-checks, rollback and the GitHub pipeline

- Pre-checks: current TLS handshake per virtual server, chain validation, key-certificate match, HA sync status
- Post-checks: new serial served on every virtual server, OCSP and chain valid, no handshake errors; automatic rollback to the previous object on failure
- GitHub pull request per renewal plan with Claude-written summary; GitHub Actions runs pytest, the dry run and the post-check report
- Scheduling renewals ahead of expiry and reporting to the certificate owner

HANDS-ON LAB
Break the chain on purpose, watch the post-check fail and the rollback restore service, then ship a clean renewal through the pull request pipeline.

DELIVERABLE
Validation, rollback and CI pipeline.

MODULE 9

Capstone: end-to-end certificate renewal across the lab estate

- Inventory in, approved and validated renewals out on F5, NetScaler, Radware and other load balancers (Anaya LB)
- No install without an approved change and a named approver; every step in the audit log
- Acceptance review, limitations statement and team demo

HANDS-ON LAB
Capstone build and demo.

DELIVERABLE
End-to-end renewal pipeline and operator runbook.

Capstone

Capstone project

A repeatable pipeline that discovers expiring certificates across F5, NetScaler, Radware and other load balancers (Anaya LB), has Claude draft a cited renewal plan and ServiceNow change, installs through deterministic tested executors only after human approval, validates every binding, rolls back on failure and leaves a complete audit trail.

Assessment

Assessment area	Weight	Evidence
Working functionality	25%	Demonstrable end-to-end workflow in a repeatable lab
Security and safety controls	20%	Threat model, authorization boundaries, approval gates, safe failure
Testing and evaluation	20%	Automated tests, golden dataset, measured quality and known limitations
Engineering quality	15%	Clear interfaces, error handling, reproducible setup, maintainable code
Evidence and explainability	10%	Traceable findings, sources, assumptions, logs
Documentation and presentation	10%	Architecture diagram, README, runbook, demo, lessons learned

Engineering principles used throughout

- Model output is a proposal, not a fact: every finding and generated change is validated by deterministic checks and tests.
- Authorization lives outside the model: tool access is enforced by application code, scoped credentials and policy.
- Human approval for consequential actions: deploying a mitigation, changing production configuration, rotating credentials.
- Decisions, evidence, model and tool versions, approvals and outcomes are logged without recording secrets.
- A model adapter and a tool adapter layer keep every build provider-agnostic and enterprise-portable.
- Quality is measured: false positives, false negatives, reliability, latency, cost and safe failure, not just a demo.

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